

POLICY BRIEF

Minimum Wage vs. Unemployment: What 30 Years of State-Level Data Show

Empirical Context for the Raise the Wage Act's \$17 Federal Minimum Wage Proposal

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Executive Summary

Our analysis of 30 years of state-level minimum wage data finds no statistically significant relationship between minimum wage increases and unemployment rates, after controlling for state and year fixed effects.

Across the two additional outcomes examined, poverty rate and median household income, the data show no significant relationship with poverty, but a statistically significant positive relationship with median household income.

The evidence suggests the employment fears driving opposition to the Raise the Wage Act [1] may be overstated, while the income effect appears real and measurable, though the poverty result remains inconclusive rather than a confirmed absence of effect. However, this analysis cannot capture hour reductions, labor force participation shifts, or regional cost-of-living variation, which limit the strength of any causal claim.

Background & Policy Context

The federal minimum wage has remained at \$7.25 per hour since 2009 [2], the longest stretch without an increase since the wage floor was established in 1938. Over the same period, a growing number of states have raised their own minimum wages well above the federal floor, creating a patchwork of wage policy across the country and a natural laboratory for studying its effects.

The Raise the Wage Act would raise the federal minimum wage to \$17 per hour by 2030 [1], reigniting a decades-old economic debate: do minimum wage increases meaningfully reduce employment, as classical labor market theory predicts [3], or do modern labor markets absorb these increases with little to no disemployment effect, as an emerging body of empirical research (following Card & Krueger, among others) suggests [4]?

The Debate

Opponents of increases above the federal floor argue that raising the price of labor reduces employer demand for it, forcing layoffs, hiring freezes, or business closures among firms operating on thin margins. They also argue that employers facing higher labor costs pass them through to consumers as higher prices, and that a higher wage floor accelerates automation, pushing employers to replace low-wage workers with technology sooner than they otherwise would have [5].

Advocates counter that higher wages lift low-income households out of poverty directly, and that the additional income low-wage workers spend circulates back into the economy as consumer demand. They also point to productivity and retention benefits: higher pay is associated with improved worker morale and lower costly turnover, and higher earnings can reduce how many workers rely on public assistance programs [6].

This brief's evidence speaks directly to only part of this debate. Its three outcome measures, unemployment, poverty, and household income, can support or challenge the job-loss and poverty-reduction claims on each side. They cannot test claims about price pass-through to consumers, automation substitution, or productivity and turnover effects, which would require different data (e.g., pricing data, capital investment, employer surveys) outside this analysis's scope.

This brief does not take a position on the proposal. Its purpose is to bring state-level evidence, accumulated over decades of natural policy variation, to bear on the empirical parts of this question, so that the debate can be grounded in data rather than priors alone.

Data & Methodology

Data Sources

All four panel variables were sourced from the Federal Reserve Economic Data (FRED) API: state-level unemployment rate (monthly, seasonally adjusted), state minimum wage (annual, sourced from the U.S. Department of Labor), median household income (annual, Census Bureau–derived), and poverty rate (annual, Census SAIPE program). Sourcing all four from a single API reduced the integration work to matching state identifiers across series, rather than reconciling different agencies' formats.

The minimum wage series reports each state's own statutory rate, which does not always reflect what is actually binding: federal law requires the higher of the state and federal rate. The **effective minimum wage** used throughout this analysis is therefore computed as the state rate or the federal rate for that year, whichever is higher. States with no minimum wage law of their own, or with a gap in their own series, default to the federal rate for the corresponding years.

Unemployment, reported monthly, was collapsed to an annual calendar-year average to match the annual frequency of the other three variables.

Panel Window: 1995–2024

The Census SAIPE program, which underlies the poverty series, has documented gaps in its early history: no state-level estimates exist for 1990, 1991, 1992, or 1994, and only isolated estimates exist for 1989 and 1993. Rather than retain an unbalanced panel around two disconnected early data points, the panel begins in 1995, where coverage becomes fully continuous across all four variables through 2024. This is a deliberate scope decision, not a limitation forced by missing data: the excluded years (1989–1994) fall outside the panel's two richest natural-experiment windows in any case, the federal minimum wage freeze at \$5.15 (1997–2007), and the freeze at \$7.25 (2009–2024), both of which see the widest state-level divergence in the sample.

The final panel spans 50 states across 30 years (1995–2024), yielding 1,500 state-year observations with no missing values and no duplicate keys.

Method

We use panel regression with state and year fixed effects. State fixed effects absorb any time-invariant differences between states (e.g., cost of living, industry mix, baseline labor market structure), while year fixed effects absorb nationwide shocks common to all states in a given year (e.g., recessions, national monetary policy). This isolates the within-state, over-time relationship between minimum wage changes and each outcome variable, which is a substantially stronger identification strategy than a simple national before/after comparison. Standard errors are clustered by state to account for serial correlation within a state's own observations over time.

This is not a randomized controlled experiment. It is an observational, quasi-experimental design that treats state-level minimum wage increases as a natural experiment, a standard and well-established approach in labor economics, but one with real limitations discussed below.

Exploratory Data Analysis

Three checks were performed before modeling. First, minimum wage trajectories across states confirm substantial, sustained policy divergence: a modest gap in the mid-1990s expands to more than \$8 an hour between states by 2024, with the widening accelerating after 2015, confirming that genuine variation exists for a fixed-effects design to exploit.

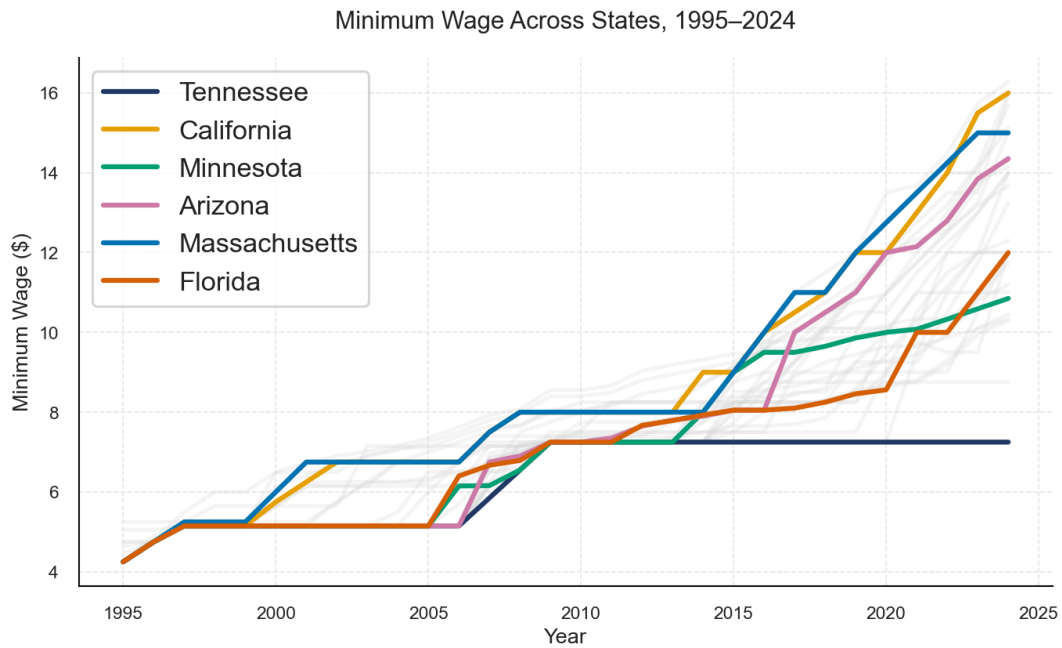


Figure 01. Minimum Wage Across States, 1995–2024

Second, unemployment trends across the same states with very different minimum wage trajectories move in near-identical shape and timing regardless of each state's own wage policy, including synchronized spikes in 2001–03, 2008–09, and 2020, visually motivating the use of year fixed effects to absorb these shared national shocks.

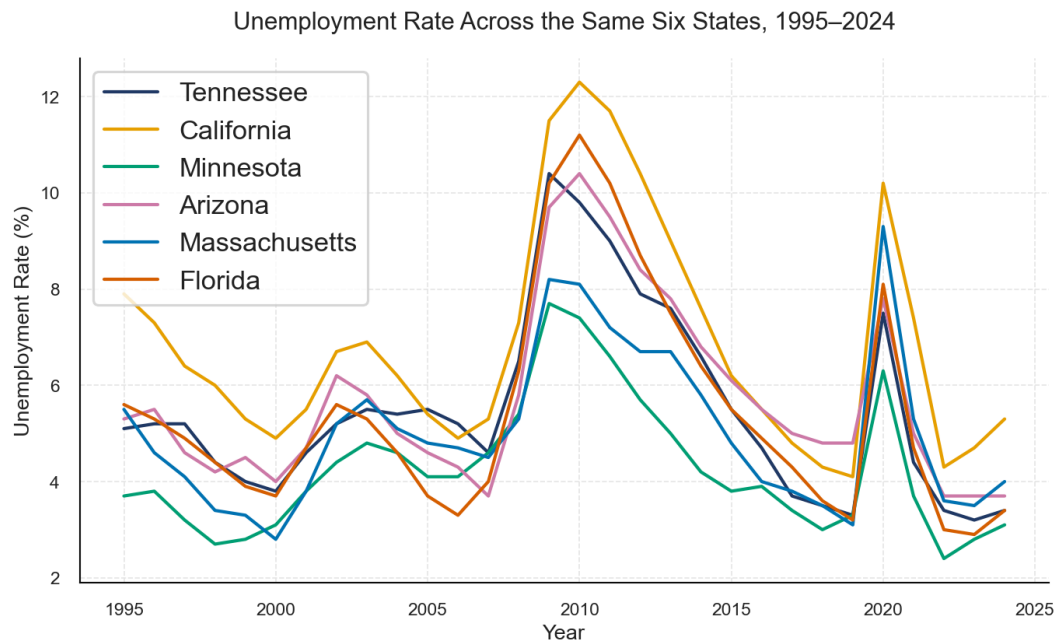


Figure 02. Unemployment Rate Across the Same Six States, 1995–2024

Third, raw pairwise correlation between minimum wage and each outcome shows a negligible relationship with unemployment and poverty, but a strong positive raw correlation with median income; the latter is treated with caution, since it likely reflects that high-minimum-wage states tend to be wealthier and more urban for reasons unrelated to wage policy itself, precisely the confound state fixed effects are designed to remove.

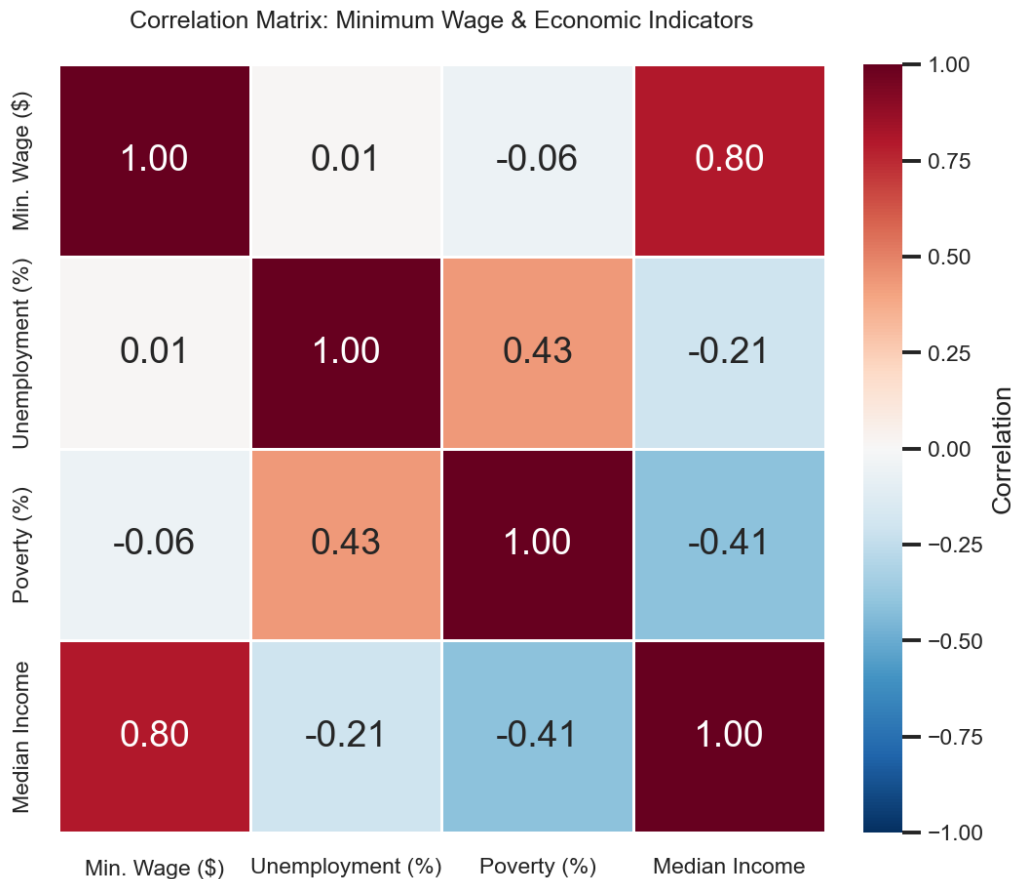


Figure 03: Correlation Matrix: Minimum Wage & Economic Indicators

Findings

Finding 1: Unemployment

We find no statistically significant relationship between minimum wage increases and state unemployment rates after controlling for state and year fixed effects (coefficient: 0.036, $p = 0.312$). This null result is consistent with recent empirical literature, following Card & Krueger, suggesting minimum wage increases have smaller disemployment effects than classical economic models predict [4].

Finding 2: Poverty Rate

States raising minimum wages above the federal floor showed no statistically significant change in poverty rates (coefficient: 0.025, $p = 0.690$), with a 95% confidence interval spanning both negative and positive values. Within this sample, minimum wage increases are not clearly associated with movement in either direction on poverty.

Finding 3: Median Household Income

Median household income showed a statistically significant positive relationship with minimum wage increases (coefficient: 1265.50, $p = 0.0001$; 95% CI: 635.53 to 1895.50). This magnitude exceeds what a single minimum-wage earner's raise alone would produce across a household, suggesting the coefficient reflects broader wage growth correlated with minimum wage policy rather than a purely mechanical pass-through effect.

The Combined Picture

Together, our three outcome measures paint a fairly consistent picture: state-level minimum wage increases above the federal floor are not associated with the unemployment increases that critics of the Raise the Wage Act predict, nor with any significant change in poverty, while showing a measurable positive relationship with household income.

If the pattern observed at the state level holds at the federal level, a significant caveat discussed below, a \$17 federal minimum wage floor would be expected to have limited impact on unemployment or poverty, while plausibly coinciding with broader income gains, though the income relationship should be treated as associative rather than a guaranteed causal lever.

Limitations & What We Cannot Conclude

- Observational data, not a controlled experiment: this design supports strong association claims but cannot prove causation.
- The unemployment rate misses hours reductions, underemployment, and labor force participation changes that a wage increase could still cause.
- State-level results may not generalize cleanly to a federal increase: a business can relocate to a lower-wage state, but no state can leave the country the way a firm can leave a high-wage state.
- Cost-of-living differences across states complicate direct comparison of nominal wage effects.
- The federal minimum wage has been stuck at \$7.25 since 2009 [2], leaving limited recent federal-level variation to study directly; all federal-level inference here is an extrapolation from state variation.

Policy Implication

The empirical evidence from state-level natural experiments suggests minimum wage increases are not associated with significant unemployment or poverty effects, while coinciding with a measurable rise in household income, which implies the job-loss concerns central to opposition against the Raise the Wage Act are not strongly supported by 30 years of state-level data. Policymakers should weigh this evidence alongside factors this analysis does not capture, such as regional cost-of-living variation, small-business margin sensitivity, and phase-in timelines.

Methodology, data sources, and full regression output available on request. This brief is an illustrative student portfolio project.

References

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